Fixed-Income Securities: Concise Summary

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List of Symbols

Notation	Description
$F_t(T_1, T_2)$	Forward rate at time t for operations between T_1 and T_2 (first argument is dropped when $t = 0$)
$f(t, T)$	Instantaneous forward rate at time t for operations at time T
$\Phi(\cdot)$	Standard normal cumulative distribution function (cdf)
$i_t(T)$	Money market (simple) rate at time t with horizon T
K_f	Forward swap rate
$\chi^2(x, y, z)$	Cumulative distribution function of the non-central chi-square law with y degrees of freedom and centrality parameter z evaluated at x
$P(t, T)$	Discount factor, zero-coupon bond price
r_t	Instantaneous risk-free rate (stochastic process)
τ	Interest rate option or swap periodicity, or Nelson-Siegel-Svensson curvature parameter
θ	Nelson-Siegel-Svensson curvature parameter
W_t	Brownian motion under the physical measure
$zcy(t, T)$	Zero-coupon yield
Z_t	Brownian motion under the Equivalent Martingale Measure

1 Fundamentals

Compounding

Discrete: $\prod_{j=0}^{n-1}(1 + r_j)$ [r_j = spot rate at beginning of period j]

Continuous: $\exp\left(\int_0^T r_u \, du\right)$ [$\{r_u\}_{u \geq 0}$ = stochastic process for instantaneous risk-free rate]

Discounting

Discrete: $\prod_{j=0}^{n-1}(1 + r_j)^{-1}$ [discount one unit from time T to current time]

Continuous: $\exp\left(-\int_0^T r_u \, du\right)$

Discount Factor (Zero-Coupon Bond Price)

$$P(t, T) = \mathbb{E}_t^Q \left[\exp\left(-\int_t^T r_u \, du\right)\right]$$

Properties: $0 < P(t,T) < 1$, decreasing and convex in T

Zero-Coupon Yield (spot rate)

$$zcy(t,T) = -\frac{1}{T-t}\ln P(t,T)$$

Forward Rates

for Simple rates: $F(T_1,T_2) = \frac{1}{T_2-T_1}\left(\frac{1+i(T_2)T_2}{1+i(T_1)T_1} - 1\right)$
for continous rates: $F(T_1,T_2) = \frac{1}{T_2-T_1}\ln\frac{P(0,T_1)}{P(0,T_2)}$
As function of zero-coupon yields: $F(T_1,T_2) = \frac{zcy(0,T_2)T_2 - zcy(0,T_1)T_1}{T_2-T_1}$

Instantaneous Forward Rate

Definition: $f(T) = \lim_{T_2\to T} F(T,T_2) = -\frac{\partial \ln P(0,T)}{\partial T}$
Relation to discount factor: $P(0,T) = \exp\left(-\int_0^T f(u)\,du\right)$

Money Market Instruments (use simple, proportional rate.)

Short maturity (below 1 year), no interim payments. eg: Treasury bills, commercial papers, banker's acceptances, municipal notes.

Discount basis: $P = M(1-i(T)T)$, yield: $y_d = \frac{M-P}{MT}$ [used in North America, UK, EU]
Money-market basis: $P = \frac{M}{1+i(T)T}$, yield: $y_m = \frac{M-P}{PT}$ [used in Japan]
Difference: money-market basis accounts for time value of interests, more favorable to lender than discount basis.

Repurchase agreements (repos): sell securities with commitment to repurchase. Financing rate implied by sell/repurchase price difference. Close to central bank overnight rate, affected by liquidity and collateral demand.

Bonds

Zero-coupon bond (STRIPS): Value $P(0,T)$ for unit face value at maturity T.
Coupon-bearing bond (par issuance, semi-annual payments):

$$\sum_{j=1}^{2T}\frac{c}{2}MP(0,\frac{j}{2}) + MP(0,T)$$

[c = coupon rate, M = face value, T = maturity]
Re-evaluation between coupon dates accounts for accrued interest.
Embedded options: callable (can be redeemed by the issuer at pre-specified price/period), convertible (can be converted by the holder into a pre-specified fraction of equity, during a given period.), callable convertible (combines call and conversion features), putable (can be redeemed by the holder at pre-specified price/period).

Forward Rate Agreement (FRA)

OTC, notional M, effective date T_1, termination date T_2, reference rate i_t, fixed rate R.
Buyer cash flow at T_1: $\frac{M(T_2-T_1)(i_{T_1}(T_2)-R)}{1+i_{T_1}(T_2)(T_2-T_1)}$ — seller cash flow is negative of buyer's.
If initial value nil: $R = F(T_1,T_2)$
No principal exchange. Flexible hedging instrument. Buyer can borrow at rate R for period $[T_1,T_2]$.

FRA Futures

Montreal Exchange - CORRA futures (1-month, 3-month maturity): quoted as $100 - R$ where R = compounded daily CORRA. Contract size: $\$2500 \times (100 - R)$ per contract [\$25 per basis point — 1 bp =0.0001].
CME group - SOFR futures (1-month, 3-month maturity): similar to CORRA. Available up to 10 years ahead.
Hedging by selling h futures (3-month, M million loan at variable rate): $h = \frac{M \text{ mil.} \times 1 \text{ bp} \times 91/365}{25} \approx M$

Treasury Bill Futures

CME group: quoted as $100 - R$ where R = yield on 13-week T-bill. Contract size: $\$2500 \times (100 - R)$ [\$25 per basis point]. Limited to one year of traded reference quarters.

Treasury Bond Futures

Underlying and Delivery: Montreal Exchange - 4 maturities (2.5, 10, 20, 30 years). Government bond with \$100,000 nominal and 6% coupon. Conversion factor (CF) adjusts delivered bond yields to 6%. Physical delivery.
Delivery Options: - Timing option: seller delivers any day in delivery month - End-of-month option: few business days between last trading day and delivery date - Quality option: seller chooses bond from pool of deliverable bonds
Invoice Price:

$$IP_t = H_{t,T} \times CF$$

where $H_{t,T}$ = futures price at sale, CF = conversion factor for delivered bond
Forward Price (Cash & Carry):

$$FP_{t,T} = (SP_t + AC_t)\left(1 + r\frac{T-t}{365}\right) - AC_T - \frac{c}{2}\left(1 + r\frac{T-s}{365}\right)$$

where SP_t = spot price at time t, AC_t = accrued interest at time t, AC_T = accrued interest at delivery T, r = implicit financing rate, c = coupon, s = last coupon date before T
Basis After Carry:

$$BAC_t = FP_{t,T} - IP_t$$

Cheapest-to-deliver (CTD) bond: minimize BAC_t at time t (Maximize r^* to find CTD. Higher r^* favors seller over holding bond.)
Implied Repo Rate:

$$r^* = \frac{H_{t,T} \times CF - SP_t + AC_T - AC_t - c/2}{(SP_t + AC_t)\frac{T-t}{365} - \frac{c}{2}\frac{T-s}{365}}$$

Options on Bond Futures

American options written on bond futures (Montreal: 2-year, 5-year, 10-year). Option maturities = 3 months preceding futures maturity. Weekly options also available.

Interest Rate Options (Caps & Floors)

OTC derivatives. *Cap* = chain of European call options ("caplets"). *Floor* = chain of European put options ("floorlets").

Contract: notional M, reference rate $i_t(\cdot)$, strike K, maturity T, settlement dates $t_1, t_2, \ldots, t_n = T$.
Period: $\tau = t_j - t_{j-1}$ (typically quarterly)
Caplet payoff at t_j (j=1,...,n):

$$M\tau(i_{t_{j-1}}(t_j) - K)^+$$

This is payoff of put on $P(t_{j-1}, t_j)$ with: strike $K' = \frac{1}{1+K\tau}$, nominal $M' = M(1 + K\tau)$, maturity t_{j-1}

$$\text{Cap}(M, K) = \sum_{j=\alpha+1}^{n} M'\text{Put}(P(t_{j-1}, t_j), K')$$

$$\text{Floor}(M, K) = \sum_{j=\alpha+1}^{n} M'\text{Call}(P(t_{j-1}, t_j), K')$$

where t_α = first reference rate reset date, $t_{\alpha+1}$ = first settlement date.

Interest Rate Swaps

Collar: Buy cap at K_2, sell floor at K_1 (where $K_1 < K_2$). Net position each quarter t_j:

$$-M\tau i_{t_{j-1}}(t_j) + M\tau(i_{t_{j-1}}(t_j) - K_2)^+ - M\tau(K_1 - i_{t_{j-1}}(t_j))^+$$

When $K_1 = K_2 = K$, collar becomes *interest rate swap.* No-arbitrage:

$$\text{Cap}(M, i, K) - \text{Floor}(M, i, K) = \text{Payer swap}(M, i, K)$$

$$\text{Floor}(M, i, K) - \text{Cap}(M, i, K) = \text{Receiver swap}(M, i, K)$$

Payer swap (fixed rate payer): pays K per period, receives floating $i_{t_{j-1}}(t_j)$

$$\text{Payer swap}(M, i, K) = M\tau \sum_{j=\alpha+1}^{n} P(0, t_j)(F(t_{j-1}, t_j) - K)$$

where $F(t_{j-1}, t_j) = \frac{1}{\tau}\left(\frac{P(0,t_{j-1})}{P(0,t_j)} - 1\right)$ = simply compounded forward rate
Simplified form:

$$\text{Payer swap}(M, i, K) = M(P(0, t_\alpha) - P(0, t_n)) - M\tau K \sum_{j=\alpha+1}^{n} P(0, t_j)$$

Receiver swap (fixed rate receiver): receives K per period, pays floating

$$\text{Receiver swap}(M, i, K) = M(P(0, t_n) - P(0, t_\alpha)) + M\tau K \sum_{j=\alpha+1}^{n} P(0, t_j)$$

Forward swap rate: Fixed rate for which payer and receiver swaps have zero value

$$K_f = \frac{P(0, t_\alpha) - P(0, t_n)}{\tau \sum_{j=\alpha+1}^{n} P(0, t_j)}$$

Swaptions

European option on interest rate swap. *Payer swaption*: right to enter payer swap. *Receiver swaption*: right to enter receiver swap. Swaption maturity $= t_\alpha$ (swap's first reset date). Tenor $= t_n - t_\alpha$.
Payer swaption payoff:

$$M\tau \sum_{j=\alpha+1}^{n} P(t_\alpha, t_j)(F_{t_\alpha}(t_{j-1}, t_j) - K)^+$$

Thus, a swaption can be viewed as an option on a coupon-bearing bond.
Compared to caps/floors: swaption value < cap/floor value since:

$$\left(\sum_{j=\alpha+1}^{n} P(t_\alpha, t_j)(F_{t_\alpha}(t_{j-1}, t_j) - K)\right)^+ \leq \sum_{j=\alpha+1}^{n} P(t_\alpha, t_j)(F_{t_\alpha}(t_{j-1}, t_j) - K)^+$$

The positive part cannot be "pushed inside" the sum.

note: January (31), February (28), March (31), April (30), May (31), June (30), July (31), August (31), September (30), October (31), November (30), December (31)

2 Yield Curve Smoothing

2.1 Money Market to Zero-Coupon Bonds

Money market instruments quoted on discount basis. For t-maturity money market rate $R_M(t)$ (day-count: Actual/365 Canada, Actual/360 US):

$$P(0, t) = 1 - R_M(t) \cdot t$$

2.2 Bootstrapping Method

Long-term swap/bond rates $R_S(T)$ (bullet redemption, semi-annual coupons):

$$100 = \sum_{i=1}^{T} \frac{100 \times R_S(T)}{2} P\left(0, \frac{i}{2}\right) + 100 \times P(0, T)$$

Solve for discount factors $P(0, i)$ recursively:
Initial: $P(0, 1/2)$ and $P(0, 1)$ from money market.
Next: From 2-year OIS rate/Bond yield $R_S(2)$, Solve for $P(0, 2)$. for $P(0, 1.5)$, linearly interpolate between $P(0, 1)$ and $P(0, 2)$. then turn the Ps into ZCYs.

2.3 Nelson-Siegel-Svensson Smoothing

$$\text{zcy}_{\text{NSS}}(T) = a + b\left(\frac{1 - e^{-T/\tau}}{T/\tau}\right) + c\left(\frac{1 - e^{-T/\tau}}{T/\tau} - e^{-T/\tau}\right) + d\left(\frac{1 - e^{-T/\theta}}{T/\theta} - e^{-T/\theta}\right)$$

Parameters: a = long-end level; b = short-end/slope; c, d = two curvatures; τ, θ = maturity of two curvatures.

$$\text{Instantaneous forward rate:} f_{\text{NSS}}(T) = a + \frac{e^{-T/\tau}}{T}(b\tau + cT) + \frac{e^{-T/\theta}}{T}dT$$

Fitting: set $w_i = 1$,for no weighting. $w_i = \frac{1/t_i}{\sum_{k=1}^{n} 1/k}$ (short-end emphasis) or $w_i = \frac{i}{\sum_{k=1}^{n} k}$ (long-end emphasis).

$$\min_{a,b,c,d,\tau,\theta} \sum_{i=1}^{n} (\text{zcy}_{\text{NSS}}(t_i) - \text{zcy}(t_i))^2 w_i^2$$

3 One-factor Models

3.1 Vasicek Model (equilibrium model)

EMM dynamics of short rate:

$$dr_t = \kappa(\theta - r_t)dt + \sigma dW_t$$

κ = speed of mean reversion; θ = long-run mean; σ = volatility. all must be positive. Negative rates possible but unlikely with reasonable parameters.

$$r_T = r_t e^{-\kappa(T-t)} + \theta(1 - e^{-\kappa(T-t)}) + \sigma \int_t^T e^{-\kappa(T-u)} dW_u$$

$$\mathbb{E}_t[r_T] = \theta + (r_t - \theta)e^{-\kappa(T-t)}$$

$$\text{Var}_t[r_T] = \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa(T-t)})$$

Physical measure:

$$dr_t = \kappa^P(\theta^P - r_t)dt + \sigma dZ_t$$

where $\kappa^P = \kappa - \sigma\lambda_1$ and $\theta^P = \frac{\kappa\theta + \sigma\lambda_0}{\kappa - \sigma\lambda_1}$.
Stationarity requires: $\lambda_1 < \kappa/\sigma$.

$$P_{vk}(t,T) = a_{vk}(t,T)\exp[-b_{vk}(t,T)r_t]$$

$$\text{zcy}_k(t,T) = -\frac{1}{T-t}\ln[a_{vk}(t,T)) - b_{vk}(t,T)r_t]$$

where

$$a_{vk}(t,T) = \exp\left[\left(\theta - \frac{\sigma^2}{2\kappa^2}\right)(b_{vk}(t,T) - T + t) - \frac{\sigma^2}{4\kappa}b_{vk}(t,T)^2\right]$$

$$b_{vk}(t,T) = \frac{1 - e^{-\kappa(T-t)}}{\kappa}$$

Option on zero-coupon bond: European call on nominal \$1 zero-coupon bond (maturity T) with strike K and option maturity T_{call}:

$$\text{call}_{vk}(P,K) = P_{vk}(0,T)\Phi(h) - KP_{vk}(0,T_{\text{call}})\Phi(h - \sigma_p)$$

$$h = \frac{\sigma_p}{2} + \frac{1}{\sigma_p}\ln\frac{P_{vk}(0,T)}{KP_{vk}(0,T_{\text{call}})}$$

$$\sigma_p = \frac{\sigma}{\kappa}(1 - e^{-\kappa(T-T_{\text{call}})})\sqrt{\frac{1 - e^{-2\kappa(T_{\text{call}}-t)}}{2\kappa}}$$

$$\text{put}_{vk}(P, K) = KP_{vk}(0, T_{\text{put}})\Phi(\sigma_p - h) - P_{vk}(0, T)\Phi(-h)$$

Interpretation (Black-Scholes analogy): Bond value is log-normally distributed. σ_p = volatility of zero-coupon bond returns; $\Phi(h)$ = call delta; $\Phi(h - \sigma_p)$ = probability (under EMM) call is exercised.
Option on coupon bond: Coupon bond replicates portfolio of zero-coupon bonds: $P_c(0, \{T_i\}) = \sum_{i=1}^n c_i P(0, T_i)$.
European call (strike K, maturity T_{call}) terminal payoff: $(\sum_{i=1}^n c_i P(T_{\text{call}}, T_i) - K)^+$.
Jamshidian decomposition: Find critical short rate r^* where coupon bond equals strike: $\sum_{i=1}^n c_i P(T_{\text{call}}, T_i, r^*) = K$. Then:

$$\text{call}(P_c(0, \{T_i\}), K) = \sum_{i=1}^n c_i \,\text{call}(P(0, T_i), K_i)$$

where $K_i = P(T_{\text{call}}, T_i, r^*)$.
Swaptions: Payer swaption: right to enter payer swap (fixed rate K, first reset t_α, settlements $t_{\alpha+1}, ..., t_n$).
Rewrite as pseudo-coupon bond: $c_i = K\tau$ for $i = \alpha + 1, ..., n - 1$; $c_n = 1 + K\tau$.
Find r^* (at-the-money): $\sum_{i=\alpha+1}^n c_i P(t_\alpha, t_i, r^*) = 1$.

$$\text{payer swaption}(M, K, t_\alpha) = M \sum_{i=\alpha+1}^n c_i \,\text{put}(P(t_\alpha, t_i), K_i)$$

$$\text{receiver swaption}(M, K, t_\alpha) = M \sum_{i=\alpha+1}^n c_i \,\text{call}(P(t_\alpha, t_i), K_i)$$

where $K_i = P(t_\alpha, t_i, r^*)$. Use zero-coupon bond options to evaluate each term.

3.2 Cox-Ingersoll-Ross (CIR) Model (equilibrium model)

EMM dynamics:

$$dr_t = \kappa(\theta - r_t)dt + \sigma\sqrt{r_t}dW_t$$

Parameters κ, θ, σ as in Vasicek. Key feature: **square-root process** ensures $r_t \geq 0$ if $r_0 \geq 0$ and Feller condition holds: $2\kappa\theta > \sigma^2$.
Distribution: r_t follows non-central chi-square (scaled). For simulation: $r_t = Y/b_t$ where $b_t = \frac{4\kappa}{\sigma^2(1-e^{-\kappa t})}$ and $Y \sim \chi^2_\nu(\lambda_c)$ non-central chi-square with $\nu = 4\kappa\theta/\sigma^2$ degrees of freedom and non-centrality $b_t r_0 e^{-\kappa t}$.
Physical measure:

$$dr_t = \kappa^P(\theta^P - r_t)dt + \sigma\sqrt{r_t}dZ_t$$

where $\kappa^P = \kappa - \sigma\lambda$ and $\theta^P = \frac{\kappa\theta}{\kappa - \sigma\lambda}$. Stationarity: $\lambda < \kappa/\sigma$.
Zero-coupon bond value (closed form):

$$P_{\text{cir}}(t, T) = a_{\text{cir}}(t, T)\exp[-b_{\text{cir}}(t, T)r_t]$$

$$a_{\text{cir}}(t,T) = \left(\frac{2\gamma e^{\frac{\kappa+\gamma}{2}(T-t)}}{(\kappa+\gamma)(e^{\gamma(T-t)}-1)+2\gamma} \right)^{\frac{2\kappa\theta}{\sigma^2}}$$

$$b_{\text{cir}}(t,T) = \frac{2(e^{\gamma(T-t)}-1)}{(\kappa+\gamma)(e^{\gamma(T-t)}-1)+2\gamma}$$

$$\gamma = \sqrt{\kappa^2 + 2\sigma^2}$$

$$\text{zcy}(t,T) = -\frac{1}{T-t}[\ln(a_{\text{cir}}(t,T)) - b_{\text{cir}}(t,T)r_t]$$

Option on zero-coupon bond (CIR): European call with strike K and maturity T_{call}:

$$\text{call}_{\text{cir}}(P,K) = P_{\text{cir}}(0,T)\chi^2_\xi(\cdot) - KP_{\text{cir}}(0,T_{\text{call}})\chi^2_\psi(\cdot)$$

where χ^2 denotes non-central chi-square distribution.

3.3 Hull-White Model (arbitrage model)

$$dr_t = \kappa(\varphi(t) - r_t)dt + \sigma dW_t$$

where $\varphi(t)$ (deterministic function of time) calibrates perfectly to the observed yield curve at $t = 0$. At $t = 0$, model fits the yield curve exactly: $P_{\text{hw}}(0,T) = P^m(0,T)$ (observed prices).
Zero-coupon bond value: Same affine structure as Vasicek:

$$P_{\text{hw}}(t,T) = a_{\text{hw}}(t,T)\exp[-b_{\text{hw}}(t,T)r_t]$$

where $b_{\text{hw}}(t,T) = \frac{1-e^{-\kappa(T-t)}}{\kappa}$ (identical to Vasicek) and

$$a_{\text{hw}}(t,T) = \frac{P^m(0,T)}{P^m(0,t)}\exp[b_{\text{hw}}(t,T)f^m(0,t) - \frac{\sigma^2}{4\kappa}b_{\text{hw}}(t,T)^2(1-e^{-2\kappa t})]$$

All other formulas follow Vasicek structure.

3.4 Brigo-Mercurio Model (arbitrage model)

Shifted CIR model: Adds deterministic shift to CIR to fit yield curve to fit at $t = 0$:

$$dx_t = \kappa(\theta - x_t)dt + \sigma\sqrt{x_t}dW_t$$

$$r_t = x_t + \varphi(t)$$

Model fits yield curve when: $\varphi(t) = f^m(0,t) - f^{cir}(0,t)$
Warning: Unlike CIR, Brigo-Mercurio does **not guarantee positive rates**. positivity is gauranteed if $\varphi(t) > 0$ for all t;
Zero-coupon bond value:

$$P_{\text{bm}}(t,T) = a_{\text{bm}}(t,T)\exp[-b_{\text{bm}}(t,T)x_t]$$

where $b_{\text{bm}}(t,T) = b_{\text{cir}}(t,T)$ and

$$a_{\text{bm}}(t,T) = \frac{P^m(0,T)}{P^m(0,t)}\frac{a_{\text{cir}}(0,t)e^{-b_{\text{cir}}(0,t)x_0}}{a_{\text{cir}}(0,T)e^{-b_{\text{cir}}(0,T)x_0}}a_{\text{cir}}(t,T)e^{\varphi(t)(T-t)}$$

All other formulas (options) follow CIR structure.

3.5 Black-Karasinski Model (arbitrage model)

Log-normal model:

$$d\ln(r_t) = \kappa(\varphi(t) - \ln(r_t))dt + \sigma dW_t$$

Key property: r_t is **log-normally distributed** and **always positive** (no negative rates). Equivalently, let $x_t = \ln(r_t)$; then x_t follows Vasicek dynamics with mean-reverting level $\varphi(t)$. Thus $r_t = \exp(x_t)$.
By Itô's lemma, dynamics of r_t:

$$\frac{dr_t}{r_t} = \left(\kappa\varphi(t) - \kappa\ln(r_t) + \frac{\sigma^2}{2}\right)dt + \sigma dW_t$$

Solution (exponential of Vasicek solution):

$$r_t = \exp\left[\ln(r_s)e^{-\kappa(t-s)} + \kappa\int_s^t \varphi(u)e^{-\kappa(t-u)}du + \sigma\int_s^t e^{-\kappa(t-u)}dW_u\right]$$

4 Multifactor Models

4.1 Two-Factor Gaussian Model

Dynamics of factors: Hull-White (1994) two-factor model (formulation by Brigo-Mercurio 2006). Two factors $\{x_t\}_{t\geq 0}$ and $\{y_t\}_{t\geq 0}$ with $x_0 = y_0 = 0$, model calibrated at $t = 0$ on yield curve:

$$dx_t = -ax_t dt + \sigma dW_{1t}$$

$$dy_t = -by_t dt + \eta dW_{2t}$$

where $\{W_{1t}\}$ and $\{W_{2t}\}$ are Brownian motions with correlation ρ. Instantaneous short rate:

$$r_t = x_t + y_t + \varphi(t)$$

Zero-coupon bond value:

$$P_{2fg}(t,T) = \frac{P^m(0,T)}{P^m(0,t)}\exp(B(t,T))$$

$$B(t,T) = \frac{1}{2}[V(t,T) - V(0,T) + V(0,t)] - \frac{1-e^{-a(T-t)}}{a}x_t - \frac{1-e^{-b(T-t)}}{b}y_t$$

European call: Same structure as one-factor model:

$$\text{call}_{2fg}(P,K) = P_{2fg}(0,T)\Phi(h) - KP_{2fg}(0,T_{\text{call}})\Phi(h-\sigma_p)$$

where $h = \frac{\sigma_p}{2} + \frac{1}{\sigma_p}\ln\frac{P_{2fg}(0,T)}{KP_{2fg}(0,T_{\text{call}})}$ and σ_p is volatility of bond returns (derived from $V(t,T)$ variance term).
Two-factor structure captures both level and slope of yield curve; better fit than one-factor models.

4.2 Fong-Vasicek Model

Two-factor extension of Vasicek: Uses rate volatility as second factor for interpretability. Dynamics under EMM:

$$dr_t = \kappa(\theta - r_t)dt + \sqrt{v_t}dZ_{1t}$$
$$dv_t = \gamma(\bar{v} - v_t)dt + \xi\sqrt{v_t}dZ_{2t}$$

where $\{Z_{1t}\}$ and $\{Z_{2t}\}$ are correlated Brownian motions with correlation ρ. First factor r_t = short rate; second factor v_t = variance of short rate.
Zero-coupon bond value: Exponential-affine structure:

$$P_{fvk}(t,T) = \exp[-b_{fvk,1}(t,T)r_t + b_{fvk,2}(t,T)v_t + b_{fvk,3}(t,T)]$$

with $b_{fvk,1}(t,T) = b_{vk}(t,T) = \frac{1-e^{-\kappa(T-t)}}{\kappa}$, while $b_{fvk,2}(t,T)$ and $b_{fvk,3}(t,T)$ involve confluent hypergeometric functions (require numerical integration; Selby-Strickland (1995) recursive algorithm available).
Second factor captures stochastic volatility of rates; improves fit relative to constant-volatility one-factor Vasicek.

4.3 Longstaff-Schwartz Model

Extension of CIR with stochastic volatility: Dynamics under EMM for short rate r_t and volatility v_t:

$$dr_t = \mu_r(r_t, v_t)dt + a\sqrt{\frac{\beta r_t - v_t}{a(\beta - a)}}dZ_{1t} + \beta\sqrt{\frac{v_t - ar_t}{\beta(\beta - a)}}dZ_{2t}$$

$$dv_t = \mu_v(r_t, v_t)dt + a^2\sqrt{\frac{\beta r_t - v_t}{a(\beta - a)}}dZ_{1t} + \beta^2\sqrt{\frac{v_t - ar_t}{\beta(\beta - a)}}dZ_{2t}$$

Key remarks: - Brownian motions $\{Z_{1t}\}$, $\{Z_{2t}\}$ are independent - Volatility constraint: $a\tau \leq v_t \leq b\tau$ - Parameter β must be sufficiently large relative to α for stochastic volatility behavior
Zero-coupon bond:

$$P_{ls}(t,T) = a_{ls,1}(t,T)^y a_{ls,2}(t,T)^z \exp[b_{ls,1}(t,T)r_t + b_{ls,2}(t,T)v_t + b_{ls,3}(t,T)]$$

European call:

$$\text{call}_{ls}(P,K) = P_{ls}(0,T)\chi^2(\cdot;\cdot) - KP_{ls}(0,T_{\text{call}})\chi^2(\cdot;\cdot)$$

4.4 Two-Factor Cox-Ingersoll-Ross Model (Brigo-Mercurio formulation)

$$dx_t = \kappa_1(\theta_1 - x_t)dt + \sigma_1\sqrt{x_t}dZ_{1t}$$
$$dy_t = \kappa_2(\theta_2 - y_t)dt + \sigma_2\sqrt{y_t}dZ_{2t}$$

where $\{Z_{1t}\}$ and $\{Z_{2t}\}$ are independent Brownian motions. Feller condition: $2\kappa_i\theta_i \geq \sigma_i^2$ for $i = 1, 2$ ensures $x_t, y_t \geq 0$.

Instantaneous risk-free rate:

$$r_t = x_t + y_t$$

Arbitrage-free version:

$$r_t = x_t + y_t + \varphi(t)$$

with shift $\varphi(t) = f^m(0,t) - f_1^{cir}(0,t) - f_2^{cir}(0,t)$. *Zero-coupon bond value:* Product of two one-factor CIR bond values:

$$P_{2fcir}(t,T) = a_{cir,1}(t,T)a_{cir,2}(t,T)\exp[-b_{cir,1}(t,T)x_t - b_{cir,2}(t,T)y_t]$$

with $a_{cir,i}(t,T) = a_{cir}(t,T)$ and $b_{cir,i}(t,T) = b_{cir}(t,T)$ from one-factor CIR model, replacing parameters $\{\kappa, \theta, \sigma\}$ with $\{\kappa_i, \theta_i, \sigma_i\}$ for $i = 1, 2$.

4.5 Multifactor Case

N-factor extension: Each factor follows CIR dynamics $dx_{it} = \kappa_i(\theta_i - x_{it})dt + \sigma_i\sqrt{x_{it}}dZ_{it}$ with independent Brownian motions; instantaneous rate $r_t = \sum_{i=1}^N x_{it}$ (or $+\varphi(t)$ for arbitrage-free version). Zero-coupon bond: product of N one-factor CIR bond values.

4.6 Affine Term Structure Models - Duffie-Kan

N exogenous factors $X_t = (X_{1t}, \ldots, X_{Nt})' \in \mathbb{R}^N$ follow under EMM:

$$dX_t = (K\theta - KX_t)dt + S_t dW_t$$

where K is $N \times N$ matrix, θ is $N \times 1$ vector, W_t is N-dimensional Brownian motion, S_t is $N \times N$ diagonal matrix with elements $[S_t]_{ii} = \sqrt{a_i + \beta_i' X_t}$ $[a_i \geq 0,\ \beta_i \in \mathbb{R}^N]$.
Market price of risk: Structure $\lambda_t = \lambda_1 S_t + S_t \lambda_2 X_t$ with $\lambda_1 \in \mathbb{R}^N$, λ_2 matrix of dimension $N \times N$.
Linear short rate:

$$r_t = \delta_0 + \delta_1' X_t$$

with $\delta_0 \in \mathbb{R}$, $\delta_1 \in \mathbb{R}^N$.
Zero-coupon bond (exponential-affine):

$$P_{aff}(t,T) = \exp[A(\tau) - B(\tau)'X_t]$$

with $\tau = T - t$ [time-to-maturity].

5 Implementation

Estimation vs Calibration: Estimation fits model to historical yield curve (ex-post analysis); calibration fits to contemporaneous market data (yields + derivatives prices) for precise point-in-time pricing. procidure: for equilibrium models, fit to yiled curve (minimizing squared errors); for arbitrage models, fit to yield curve then fit to caps/floors or swaption prices. we check the price fit but a more stringent test is to check the implied volatilities (using Blacks) fit.

5.1 Black Model for Caps and Floors

Black model approximation: Assumes forward rates are lognormal; ignores correlation between discount factor and payoff; recognizes future spot rate $\approx$ future forward rate $l_{t_{j-1}}(t_j) = F(t_{j-1}, t_j)$.
Cap price with strike K and maturity t_n:

$$\text{cap}(t_n) = M \sum_{j=i+1}^{n} \tau P(0, t_j) \left[F(t_{j-1}, t_j)\Phi(d_1) - K\Phi(d_2) \right]$$

where τ is accrual period (typically 0.25 for quarterly), M is notional, $P(0, t_j)$ is discount factor, and:

$$d_1 = \frac{1}{v_j} \left[\ln \frac{F(t_{j-1}, t_j)}{K} + \frac{v_j^2}{2} \right], \quad d_2 = d_1 - v_j, \quad v_j = \sigma_{\text{cap}} \sqrt{t_{j-1}}$$

with σ_{cap} = implied volatility from market quotes (typically varies by tenor; first reset at $t_0 = 0.25$, $t_1 = 0.5$, etc.).
Floor price: By put-call parity:

$$\text{floor}(t_n) = M \sum_{j=i+1}^{n} \tau P(0, t_j) \left[K\Phi(-d_2) - F(t_{j-1}, t_j)\Phi(-d_1) \right]$$

At-the-money (ATM) strike: Cap and floor are ATM when $K = K_f$ (forward swap rate):

$$K_f = \frac{P(0, t_a) - P(0, t_n)}{\sum_{j=i+1}^{n} \tau P(0, t_j)}$$

5.2 Black Model for Swaptions

Black approximation: Ignores correlation between discount factor and swaption payoff; assumes forward swap rates are driftless geometric Brownian motions under EMM.
Payer swaption price with strike K and maturity t_n:

$$\text{payer swaption}(t_n) = M \left(K_f \Phi(d_1) - K\Phi(d_2) \right) \sum_{j=a+1}^{n} \tau P(0, t_j)$$

Receiver swaption price:

$$\text{receiver swaption}(t_n) = M \left(K\Phi(-d_2) - K_f \Phi(-d_1) \right) \sum_{j=a+1}^{n} \tau P(0, t_j)$$

where K_f is forward swap rate, and:

$$d_1 = \frac{1}{v} \left[\ln \frac{K_f}{K} + \frac{v^2}{2} \right], \quad d_2 = \frac{1}{v} \left[\ln \frac{K_f}{K} - \frac{v^2}{2} \right]$$

$$v = \sigma_{\text{swapt}} \sqrt{t_a}$$

with σ_{swapt} = implied swaption volatility from market, t_a = start of swap period.

5.3 Kalman Filtering

State-space framework: Measurement equation relates observed zero-coupon yields to hidden state variables; transition equation describes state variable evolution; estimates parameters maximizing total log-likelihood.
Measurement equation:

$$ZCY_t = A + BX_t + \eta_t$$

where A is $N \times 1$ vector, B is $N \times K$ matrix, η_t is $N \times 1$ vector of measurement errors with diagonal covariance matrix $H = \text{diag}(h_{t_1}^2, \ldots, h_{t_N}^2)$ [variances may differ by maturity but constant over time; uncorrelated cross-sectionally].
Transition equation:

$$x_{i,t} = \mathbb{E}[x_{i,t}|x_{i,t-\delta}] + \sqrt{\text{Var}[x_{i,t}|x_{i,t-\delta}]}\xi_t$$

Algorithm: Loop over all observation dates: (1) predict next-period state estimators and error variance; (2) compute prediction error (observed yields − model-predicted yields) and its variance; (3) compute log-likelihood from prediction error; (4) update state estimators using prediction error and updated variance. Final parameters maximize total log-likelihood over sample.

5.4 Diebold-Li Approach

Dynamic extension of Nelson-Siegel smoothing.
with time-varying loadings $L_a = 1$, $L_b = \frac{1-e^{-\tau/\tau}}{T}$, $L_c = \left(\frac{1-e^{-\tau/\tau}}{T} - e^{-\tau/T}\right)$ [parameter τ fixed]. Factor representation:

$$zcy_t(T) = a_t L_a + b_t L_b + c_t L_c$$

where a_t = level, b_t = slope, c_t = curvature.
Forecasting: Set parameter τ such that curvature loading L_c reaches maximum at 30 months. Specify AR(1) dynamics for each factor:

$$\hat{x}_{t+h} = c_x + k_x \hat{x}_t, \quad \text{for } x = \{a, b, c\}$$

Parameter estimation: OLS regression of x_{t+h} on x_t estimates intercept c_x and slope k_x by minimizing squared residuals between observed future factor values and predicted values.

6 HJM and Market Models

HJM and Market Models

HJM General Framework

Heath, Jarrow & Morton (1992) model the entire yield curve by working directly with **instantaneous forward rates** $f(t, T)$ rather than the short rate r_t. The key insight: specifying the **forward rate volatility function** $\sigma(t, T)$ fully determines the dynamics of the whole yield curve. This opens a general family of term structure models.

Core result (HJM drift condition): Under the EMM, the drift of the instantaneous forward rate is *entirely pinned down* by its volatility — no free parameters remain for the drift. Formally, the forward rate dynamics are:

$$df(t,T) = \sigma(t,T) \int_t^T \sigma(t,u)\, du\, dt + \sigma(t,T)\, dW_t$$

Specifying $\sigma(t,T)$ completely characterizes the model for all maturities simultaneously.

Markov condition (Carverhill 1994): In general, $r_t = f(t,t)$ is *not* Markovian, making tree implementation impossible. The short rate is Markovian if and only if $\sigma(t,T) = h(t) \cdot g(T)$ (separable form). In that case the HJM model reduces to a **generalized Hull-White model** and can be implemented on a recombining tree.

Volatility specifications:

- **Ritchken–Sankarasubramanian**: $\sigma(t,T) = \psi_t \cdot \exp(-\kappa(T-t))$ with ψ_t stochastic; leads to a two-state Markov representation (r_t, ϕ_t) enabling tree/PDE implementation.
- **Li–Ritchken–Sankarasubramanian**: special case with $\sigma(t,T) = v_t^\beta \exp(-\kappa(T-t))$, $0 \le \beta \le 1$; interpolates between Gaussian ($\beta = 0$) and log-normal ($\beta = 1$) dynamics.
- **Mercurio–Moraleda / Moraleda–Vorst**: polynomial-exponential forms that generate a **humped term structure of volatility** (volatility peaks at intermediate maturities) — a stylized fact of cap implied vol surfaces. Former lacks closed-form ZCB; latter has closed-form ZCB.

Market Models (BGM/LMM)

Motivation: Market quotes cap/floor/swaption prices as Black implied vols, but the Black formula rests on invalid assumptions (ignores discount factor–payoff correlation; assumes driftless GBM for forward rates under the EMM). Market models *rigorously justify* the Black formula via a **change of probability measure**.

Key mechanism: The simply-compounded forward rate $F(t;T_1,T_2)$ is a ratio of traded assets divided by $P(t,T_2)$, so it is a T_2**-forward martingale** under the T_2-forward measure $\mathbb{Q}^{T_2}$. Under this measure it can be modeled as a driftless GBM, exactly recovering the Black caplet formula without approximations.

Cap Market Model (BGM/Miltersen–Sandmann–Sondermann/Jamshidian 1997):

- Each forward rate $F(t;T_{k-1},T_k)$ follows driftless GBM under its own forward measure $\mathbb{Q}^{T_k}$.
- Caplets are priced exactly with the Black formula; cap = sum of Black caplets.
- Calibration: fit the volatility function $\gamma(t,T_k)$ (e.g., parametric humped form $\gamma = [\alpha+\beta(T_k-t)]e^{-\delta(T_k-t)}+\theta$) to market cap implied vols by minimizing squared pricing errors.
- **Swaptions cannot be priced analytically** in the cap market model $\Rightarrow$ use Monte Carlo simulation.
- Under a different forward measure $\mathbb{Q}^{T_n}$, forward rates acquire a **drift correction**; simulation uses this drift-adjusted dynamics via Euler scheme.

Swap Market Model (Jamshidian 1997):

- The forward swap rate $K_f(t)$ is a ratio of traded assets, hence a martingale under the **swap measure** (annuity as numéraire). Models $K_f(t)$ as driftless GBM $\Rightarrow$ Black swaption formula is exact.

- **Cap/floor market model and swap market model are theoretically incompatible**: one cannot simultaneously use Black formula for both caps and swaptions in a single consistent model. Practitioners must choose one and price the other numerically.

Backward-Looking Rates (Post-LIBOR, Lyashenko–Mercurio 2019)

With the end of LIBOR, caps/floors/swaptions reference **compounded overnight rates set in arrears** (SOFR, CORRA). These are **backward-looking**: the rate for period $[T_1, T_2]$ is not known until T_2.

Key idea: Extend the zero-coupon bond definition beyond maturity: for $t > T$, set $P(t,T) = B(t)/B(T)$ (bank account ratio), which remains a valid numéraire. The backward-looking forward rate $\tilde{F}(t; T_1, T_2)$ is still a T_2-martingale, giving a formula analogous to the forward-looking case.

Volatility adjustment: Before T_1, $\tilde{F}(t; T_1, T_2)$ behaves like the forward-looking rate (same volatility $\gamma(t, T_2)$). Between T_1 and T_2, compounding of overnight rates induces a **linear volatility decay**: $\bar{\gamma}(t; T_1, T_2) = \min\left(1, \frac{T_2 - t}{T_2 - T_1}\right) \gamma(t, T_2)$. Market model structure and measure-change drift corrections carry over unchanged.

7 Numerical Methods

7.1 Binomial Trees

Heston (1995) binomial tree: Discrete-time arbitrage-free tree for one-factor model where short rate follows random walk with mean reversion. Short rate dynamics:

$$r_{t+\Delta t} = \alpha + \rho r_t + \sigma \varepsilon$$

Connection to continuous models: Equivalent to Vasicek or CIR with $\kappa = 1 - \rho$ and $\theta = \alpha/\kappa$.
No-arbitrage valuation: Portfolio value equals discounted expected contingent claim under EMM:

$$g(t) = e^{-r_t \Delta t}[\pi g_u(t + \Delta t) + (1 - \pi) g_d(t + \Delta t)]$$

with risk-neutral probability:

$$\pi = \frac{e^{r_t \Delta t} - d}{u - d}, \quad u = \exp[(-\alpha - \rho r_t - \sigma)\Delta t], \quad d = \exp[(-\alpha - \rho r_t + \sigma)\Delta t]$$

Ensures martingale property: discounted value process is martingale under EMM.

7.2 Trinomial Trees

Hull-White trinomial tree: Captures richer interest rate dynamics with faster convergence to continuous time limit; more complex than binomial trees but better for mean-reverting processes. Designed for Hull-White model (can adapt for log-normal processes like Black-Karasinski). Underlying process:

$$dx_t = -\kappa x_t dt + \sigma dW_t$$

Process is normally distributed and mean-reverting to zero.

Tree structure: At each time step Δt, discretized process value $x_{i,j}$ (where i = time index, j = space index) branches to three nodes with transition probabilities π_u, π_m, π_d: - Up state: $x_{i+1,j+1}$ - Middle state: $x_{i+1,j}$ - Down state: $x_{i+1,j-1}$
Probability constraints and moment matching: Transition probabilities sum to one and match first two moments of underlying process:

$$\pi_u + \pi_m + \pi_d = 1$$

$$\pi_u(x_{i,j} + \Delta x_j) + \pi_m x_{i,j} + \pi_d(x_{i,j} - \Delta x_j) = x_{i,j} e^{-\kappa \Delta t}$$

Variance equation:

$$\pi_u(x_{i,j} + \Delta x_j)^2 + \pi_m x_{i,j}^2 + \pi_d(x_{i,j} - \Delta x_j)^2 = [x_{i,j} + \Delta x_j(\pi_u - \pi_d)]^2 + \sigma^2 \frac{1 - e^{-2\kappa \Delta t}}{2\kappa}$$

Branching geometry: Tree geometry adapts in different regions to improve mean reversion capture and reduce negative rates: - **Process (A)**: Normal branching when $x_{i,j}$ neither too high nor too low; states $\{x_{i,j} - \Delta x, x_{i,j}, x_{i,j} + \Delta x\}$ - **Process (B)**: When $x_{i,j}$ is too high (pulls down); states $\{x_{i,j} - 2\Delta x, x_{i,j} - \Delta x, x_{i,j}\}$ - **Process (C)**: When $x_{i,j}$ is too low (pulls up); states $\{x_{i,j}, x_{i,j} + \Delta x, x_{i,j} + 2\Delta x\}$
Critical levels $j_{\min}$ and $j_{\max}$ determine where branching switches from (A) to (C) and from (A) to (B). Numbering states from bottom: $x_{i,j} = j\Delta x$; offsets chosen to ensure mean reversion and positivity.

7.3 Second Stage of the Tree: Calibration via Arrow-Debreu Prices

Objective: Calibrate tree-built nodes to match market zero-coupon bond prices using Arrow-Debreu prices Q_{ij} (state prices: value of claim paying \$1 at node j, time i).
Zero-coupon bond pricing constraint with shift parameter:

$$P_{NSS}(0, (i+1)\Delta t) = \sum_k Q_{ik} \exp(-\alpha_i \Delta t) \pi_k$$

Solve for shift parameter α_i to match observed bond price $P_{NSS}(0, (i+1)\Delta t)$ from market data.
State price propagation:

$$Q_{(i+1)j} = \sum_k Q_{ik} \pi_{kj} \exp\left(-(\alpha_i + k\Delta x)\Delta t\right)$$

where π_{kj} is transition probability from state k to state j.
Black-Karasinski extension: For log-normal rates, apply $f^{-1}(\cdot) = \exp(\cdot)$ and solve α_i numerically as the equation becomes nonlinear.
Key points: Match each bond maturity sequentially; one α_i per time step (one-factor); ensure positivity of rates if using normal models; use Newton-Raphson for implicit solutions.

7.4 Finite Difference Methods

Flexibility vs. convergence trade-off: The finite difference method is more flexible than trees because it can easily accommodate a wide variety of interest rate models. However, the method is slower to converge (at least with its simplest scheme) notably because it requires the discretization of an entire state-space.

7.4.1 The One-dimensional Explicit Scheme

One-factor diffusion model under EMM:

$$dr_t = \mu(t, r_t)dt + \sigma(t, r_t)dW_t$$

PDE for security value: In the absence of arbitrage, any fixed-income security can be viewed as a contingent claim on the instantaneous risk-free rate; its value $g(t, r_t)$ satisfies:

$$\frac{\partial g(t, r_t)}{\partial t} + \frac{\partial g(t, r_t)}{\partial r_t}\mu(t, r_t) + \frac{1}{2}\frac{\partial^2 g(t, r_t)}{\partial r_t^2}\sigma(t, r_t)^2 - r_t g(t, r_t) = 0$$

Grid discretization: A finite difference method consists in discretizing this PDE in a state-space represented by a grid with space step Δr and time step Δt. State nodes indexed by i (space, from bottom to top) and t (time, from current to maturity T).
Discretized Greeks: Let Δ_i^t, Γ_i^t and θ_i^t denote the discretized versions of the contingent claim's delta, gamma and theta respectively, all being evaluated at space node i (numbered from bottom to top) and time node t (numbered from current time 0 till maturity T). The drift $\mu(t, r_t)$ and volatility $\sigma(t, r_t)$ being functions of t and r_t, their discretized versions at space node i and time node t can be written as μ_i^t and σ_i^t. Thus, the discretized PDE is:

$$\theta_i^t + \Delta_i^t \mu_i^t + \frac{1}{2}\Gamma_i^t(\sigma_i^t)^2 - i\Delta r g = 0$$

Central difference approximations:

$$\Delta_i^t = \frac{g_{i+1}^t - g_{i-1}^t}{2\Delta r}$$

$$\Gamma_i^t = \frac{g_{i+1}^t - 2g_i^t + g_{i-1}^t}{(\Delta r)^2}$$

$$\theta_i^t = \frac{g_i^t - g_i^{t-1}}{\Delta t}$$

Backward recursion formula: Plugging the above approximations into the discretized PDE and rearranging:

$$g_i^{t-1} = A_i^t g_{i-1}^{t-1} + B_i^t g_i^{t-1} + C_i^t g_{i+1}^{t-1}$$

with $v_1 = \frac{\Delta t}{(\Delta r)^2}$, $v_2 = \frac{\Delta t}{\Delta r}$, and coefficients:

$$A_i^t = \frac{1}{2}[(\sigma_i^t)^2 v_1 - \mu_i^t v_2]$$

$$B_i^t = 1 - (\sigma_i^t)^2 v_1 - i\Delta r \Delta t$$

$$C_i^t = \frac{1}{2}[\mu_i^t v_2 + (\sigma_i^t)^2 v_1]$$

Key properties and stability:

- Sum of coefficients: $A_i^t + B_i^t + C_i^t = 1 - i\Delta r \Delta t$ captures the one-period discount factor; equation (7.25) embeds the martingale property of discounted price process.
- Weights A_i^t, B_i^t, C_i^t are not probabilities and may lie outside $[0, 1]$, potentially causing instability. Stability requires small v_1 and v_2: discretize time sufficiently relative to space (horizontally stretched grid).

- Choice of $r_{\max}$: balance between avoiding truncation bias (need $r_{\max}$ sufficiently high) and computational efficiency (higher $r_{\max}$ requires finer discretization). Trade-off between accuracy and cost.

Boundary conditions:

- Zero-coupon bonds: $g = 1$ when $r = 0$; $g = 0$ when $r = r_{\max}$.
- Deep in-the-money regions: linear approximation $g^t_{i_{\max}} = 2g^t_{i_{\max}-1} - g^t_{i_{\max}-2}$.
- Deep out-of-the-money regions: derivatives are worthless.
- American options: compute maximum of exercise and continuation values recursively at each node; use exercise value as boundary condition for deep in-the-money regions.

7.4.2 Two-dimensional Explicit Scheme

Two-factor diffusion model under EMM:

$$dx_s = \mu(t, x_s, y_t)dt + \sigma(t, x_s, y_t)dW^x_s$$

$$dy_t = \mu(t, x_s, y_t)dt + \sigma(t, x_s, y_t)dW^y_s$$

with $dW^x_s dW^y_s = \rho dt$ (correlation ρ between factors). Contingent claim value $g(t, x_s, y_t)$ satisfies:

$$\frac{\partial g(t,x_s,y_t)}{\partial t} + \frac{\partial g(t,x_s,y_t)}{\partial x_s}\mu(t,x_s,y_t) + \frac{1}{2}\frac{\partial^2 g(t,x_s,y_t)}{\partial x_s^2}\sigma(t,x_s,y_t)^2 + \frac{\partial g(t,x_s,y_t)}{\partial y_t}\mu(t,x_s,y_t) + \frac{1}{2}\frac{\partial^2 g(t,x_s,y_t)}{\partial y_t^2}\sigma$$

3-dimensional grid discretization: Space steps Δx and Δy; time step Δt. Notation: $g^t_{i,j}$ = contingent claim value at node i for state variable x (left to right), node j for state variable y (bottom to top), at time t.

Discretization approach: Apply central differences for partial derivatives and backward recursion as in one-dimensional case. Cross-partial derivative $\frac{\partial^2 g}{\partial x_s \partial y_t}$ discretized using four-point stencil. Solution proceeds backward through time on 2-D grid at each time step.

7.5 Monte Carlo Simulation

Generating normal random variables: Box-Muller method: given u_1, u_2 from uniform $[0, 1]$:

$$\varepsilon_1 = \sqrt{-2\ln(u_1)}\cos(2\pi u_2), \quad \varepsilon_2 = \sqrt{-2\ln(u_1)}\sin(2\pi u_2)$$

Euler discretization scheme:

$$r_{t+\Delta t} = r_t + \kappa(\theta - r_t)\Delta t + \sigma\sqrt{\Delta t}\varepsilon_t$$

Full truncation scheme: To preserve non-negativity, replace r_t with $\max(r_t, 0)$ in drift and diffusion coefficients.

Milstein scheme: Higher accuracy via:

$$r_{t+\Delta t} = r_t + \kappa(\theta - r_t)\Delta t + \sigma\sqrt{\Delta t}\varepsilon_t + \frac{\sigma}{4}\Delta t(\varepsilon_t^2 - 1)$$

Path-dependent valuation: For each simulated path j, compute discount factor as:

$$D^{(j)} = \exp\left(-\sum_{n=0}^{N-1} r_n^{(j)} \Delta t\right)$$

Then price security as:

$$\text{Price} = \mathbb{E}^Q[D^{(j)} \cdot X(T)] = \frac{1}{M}\sum_{j=1}^{M} D^{(j)} \cdot X^{(j)}(T)$$

where $X(T)$ is terminal payoff and M is number of paths.
Correlated Brownian motions: Generate independent $(\varepsilon_1, \varepsilon_2)$, then:

$$\Delta W_1 = \sqrt{\Delta t}\varepsilon_1, \quad \Delta W_2 = \sqrt{\Delta t}(\rho\varepsilon_1 + \sqrt{1-\rho^2}\varepsilon_2)$$

7.5.1 Path-Dependent Interest Rate Payoffs

Barrier caps/floors: Activated ("in"-type) or deactivated ("out"-type) first time reference rate reaches critical barrier level from below ("up") or above ("down"). Four main types: up-in cap, down-in floor, down-out cap, up-out floor. Payoff:

$$\left(i_{u-1}^{(u)} - K\right)^+ \cdot \mathbb{1}\left(\max_{s\in[u-1,u]} i_s^{(u)} > H\right)$$

Barrier condition applies to whole cap/floor: once a caplet activates, all subsequent caplets activate.
Bounded caps/floors: Total accumulated payout cannot exceed amount M. At each settlement, accumulated value is computed and test applied to determine if below limit:

$$\left(i_{u-1}^{(u)} - K\right)^+ \cdot \mathbb{1}\left(\sum_{j=1}^{u-1}\left(i_{j-1}^{(j)} - K\right)^+ < M\right)$$

Option deactivated first time test fails.
Incremental fixed swaps: Fixed leg calculated as weighted average of fixed and variable rates. For payer swap with notional M, fixed rate K, frequency τ:

$$-M\tau\left(\kappa + (1-x)u_t^{(u)}\right) + M\tau d_t^{(u)}$$

Fraction x contractually determined by reference rate tier: $x = 100\%$ if $i_t < i_1$; $x = 60\%$ if $i_2 \leq i_t \leq i_1$; $x = 0\%$ if $i_t > i_2$. At each reset, simulated rate determines fraction x and payoff.
Ratchet caps/floors: Strike adjusts to level of reference rate on preceding reset date. Payoff at settlement:

$$\left(i_{u-1}^{(u)} - i_{u-2}^{(u-1)}\right)^+$$

Simulation adjusted to compute difference between current and previous reset rates at each settlement date.
Rolling caps/floors: Nominal is not constant. When caplet (floorlet) not exercised (not in-the-money), its notional is added to next caplet (floorlet). At each reset date, apply test and adjust payoff accordingly.

8 Credit Risk Fundamentals

8.1 Key Concepts and Stylized Facts

8.1.1 Credit Risk and Credit Spreads

Credit risk is the possibility that a counterparty fails to meet contractual obligations, causing **price fluctuations reflecting exposure to credit risk**. Exposure has two components: **timing of default** (measured by default probability) and **magnitude of default** (measured by loss upon default), decomposing as: (credit exposure) $\times$ (loss given default).
Credit risk exposure on fixed-income securities is compensated by the **credit spread**. For a debt claim with value D promising M at period end:

$$D = \frac{M}{1+y} \quad (8.1)$$

If defaultable with EMM probability q of delivering $M(1-L)$, where L is **loss given default**:

$$D = \frac{M(1-q) + M(1-L)q}{1+r} \quad (8.2)$$

where r is the risk-free rate. Combining (8.1) and (8.2):

$$y - r \approx qL \quad (8.3)$$

Fundamental decomposition: credit spread = (default probability) $\times$ (loss given default).
Yield spread vs. credit spread: Yield spread of defaultable instruments includes credit spread plus other premiums (tax, liquidity); attributing the fraction to credit risk is empirical.

8.1.2 The Term Structure of Credit Spreads

Consider homogeneous defaultable securities (comparable credit risk source) with observable market prices at different maturities. Assuming credit risk is the only factor in total yield spread, the **term structure of credit spreads (TSCS)** is calculated as the difference between yield-to-maturity and (risk-free) zero-coupon yield for the same horizon.
TSCS dynamics: Credit spreads are volatile and counter-cyclical (spikes during 2007-08 crisis and 2020 Covid pandemic). Like risk-free zero-coupon yields, the TSCS is usually increasing during quiet times and inverted in crisis times.

8.1.3 Recovery Rates

Recovery rate: One minus loss given default; one of two major constituents of credit spread. For defaultable zero-coupon bond value $D(t,T)$ (promising \$1 at T), let τ = random default time and δ = recovery rate.
Three recovery rate definitions exist:

Recovery of Face Value (RFV):
Definition: Fraction of defaultable claim nominal at default time.

Formula: $D(T,T) = 1_{\tau>T} + \frac{\delta}{P(t,T)} 1_{\tau \le T}$ (8.4)
Usage: Rating agencies (e.g., Moody's) for historical recovery rates.
Pro: Eases comparison across claims; accommodates delay to resolve default.
Con: Makes δ sensitive to timing of default.

Recovery of Treasury Value (RTV):
Definition: Fraction of equivalent risk-free zero-coupon bond at default time.
Formula: $D(T,T) = 1_{\tau>T} + \delta 1_{\tau \le T}$ (8.5)
Pro: Makes δ insensitive to timing of default; simplifies valuation of defaultable claims.
Con: Not how recovery is determined in practice.

Recovery of Market Value (RMV):
Definition: Fraction of pre-default bond value at default time. δ = proportional value loss incurred immediately by claimholder.
Formula: $D(T,T) = 1_{\tau>T} + \delta \frac{D(\tau^-,T)}{P(t,T)} 1_{\tau \le T}$ (8.6)
Theory (Duffie-Singleton 1999): Under reduced-form approach, RMV allows expressing defaultable claim value as EMM expectation at risk-adjusted rate $R_t = r_t + s_t$ (where s_t = credit spread):

$$D(t,T) = \mathbb{E}^Q \left[\exp\left(-\int_t^T (r_u + s_u)\, du \right) \right] \quad (8.7)$$

Pricing implication: Valuation of defaultable claims uses same tools as yield curve and zero-coupon bond pricing (factor models or HJM framework) for dynamics of $\{r_t\}$ and $\{s_t\}$.

Stylized Facts on Recovery Rates:
Priority impact: Recovery rates greatly impacted by claim priority; higher for secured bank loans than secured bonds.
Negative correlation: Strong stylized fact—negative correlation between default likelihood and recovery. OLS regression (Moody's 1983–2023 data): coefficient -6.56 (p-value 2.8×10^{-8}), $R^2 = 55\%$.
Amplification effect: Credit spread = default probability $\times$ loss given default (equation 8.3). Negative correlation between these factors amplifies credit spread variations.